Autonomous Video Transmission and Air-to-Ground Coordination in UAV-Swarm-Aided Disaster Response Platform

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Abstract. Unmanned Aerial Vehicles (UAVs), or drones with cameras, are crucial for environmental awareness in applications like smart agriculture, border security, and disaster response. Building realistic UAV testbeds for novel network control algorithms is challenging due to time constraints and regulatory limitations. To develop an autonomous air-to-ground coordination platform, in this paper, we introduce a UAV-swarm-aided Disaster Response Platform (DRP) that simulates video transmission and coordination that integrates simulation for both drones and networks, allowing experimentation with protocols (HTTP/TCP, UDP/RTP, QUIC) and video properties (codec, resolution). Our design combined human-centered interaction principles, prioritizing user experience and indicating multi-modal interactions. In addition, our approach utilizes trace-based experiments to demonstrate its effectiveness in delivering video quality (e.g., PSNR) aligning with real-world measurements, while in the meantime validating model accuracy. Evaluation results demonstrate that our intelligent video transmission and air-to-ground coordination strategies present reasonable video quality under measurement of subjective and objective metrics, and in the meantime achieve $\approx 85\%$ accuracy with dynamic decision-making with competitive time efficiency and energy savings ($\approx 18\%$ gain) compared to on-boarding only. Implementing these strategies in UAV-swarm-aided DRPs significantly enhances overall response efficiency, ensuring the safety of lives and property.

Keywords: Drone-Ground Interaction Platform · Disaster Response Platform · Video Transmission · Air-to-ground Coordination

1 Introduction

Unmanned Aerial Vehicles (UAVs) play a crucial role in diverse applications, including smart farming, public safety, emergency medical services, and package

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delivery. These drones empower computer systems, including Ground Control Stations (GCS), to actively manage their location, leading to coordinated efforts that significantly enhance environmental situational awareness [12]. As illustrated in Fig. 1, these systems commonly incorporate air-to-ground wireless links, establishing connections between the edge network, Unmanned Aerial Vehicles (UAVs), and Ground Control Stations (GCS). Compared to a single-UAV operation, a UAV-swarm-aided operation shows the efficiency and robustness of the system [2]. The networking architecture typically involves setting up peer-to-peer or centralized-control communication networks with two primary modes including i) Air-to-Air, and ii) Air-to-Ground.

In addition, the application of geospatial video analytics with UAV-swarm in various domains is widespread, involving intricate data processing pipelines to extract environmental situational awareness from videos captured from diverse perspectives. However, the deployment of realistic UAV swarm testbeds with appropriate network edge-communication protocols is currently a time-consuming and costly endeavor. This challenge arises from constraints such as ensuring safe operation, managing limited energy resources, and navigating government regulation restrictions [13]. Furthermore, constructing UAV swarm testbeds for high-scale experiments poses additional difficulties, particularly in creating repeatable and reproducible experiments for testing major hypotheses. The complexities involved in achieving these objectives further contribute to the overall challenges in advancing geospatial video analytics with UAV technology [16].

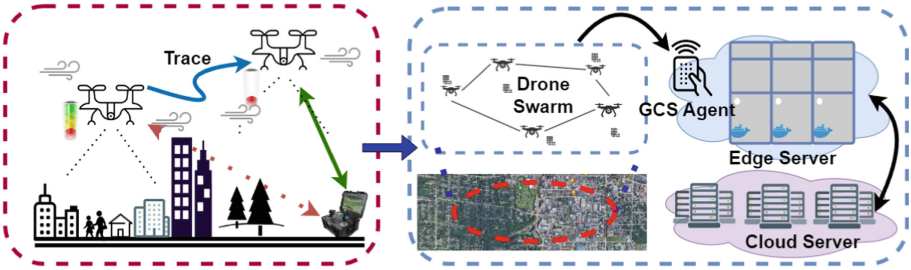


Fig. 1. Overview of converting from single UAV operation into UAV-swarm aided operation system with Air-to-Ground Coordination.

Traces derived from UAVs during video analytics simulations, offer real-time opportunities for experimentation with edge-cloud configurations, diverse UAV mobility models, and control connection status between devices and infrastructure. These traces enable adaptation and policy experiments in video analytics based on network performance measurements, incorporating Human-Computer Interaction (HCI) concepts to enhance user experience and system usability [8]. In the realm of UAV-swarm-aided geospatial video analytics algorithms, network simulations involve setting up multi-UAV configurations with custom wireless communication and network links (air-to-air; air-to-ground), along with mobility routing protocols. Moreover, despite the availability of effective UAV simulations for measuring flight movements, battery life, and application performance,

they often lack integration with relevant network simulators. Conversely, powerful network simulators exist but lack the necessary features to realistically address UAV-specific issues at a high scale [1,11]. The joint design of UAV and network simulations, incorporating HCI principles, offers significant benefits to experimenters. It allows for testing various system configurations, such as network security protocols [9], and implementing different air-to-ground coordination strategies before actual deployment in the field. This synergistic approach enhances the efficiency and effectiveness of experiments in UAV-swarm-aided configuration testing, ensuring a user-centric and user-friendly system design.

This paper investigates state-of-the-art approaches to advanced UAV-swarm-aided Disaster response Platforms (DRPs), focusing on trace generation for (a) network communication between UAV-swarm edge-server architectures and (b) computation in UAV video analytics across multiple application scenarios. Addressing knowledge gaps identified in the literature survey, we propose a trace-based DRP design utilizing NS-3 [6]. Our design integrates UAV-swarm-aided simulations within a hierarchical UAV configuration, covering wide-area search drones and intelligence-gathering drones, to serve applications requiring geospatial environmental situational awareness, with a thoughtful consideration of Human-Computer Interaction (HCI) concepts.

The hierarchical UAV platform involves collaboration between low-cost and high-cost drones using a geo-location service to provide a 'common operating picture' for decision-makers. Our innovative approach synergizes network and drone simulators/emulators, demonstrating apparent benefits in trace-based simulations involving realistic drone video analytics pipelines. Videos and traces generated from our platform can integrate various drone configurations, wireless communication links (air-to-air; air-to-ground), and mobility routing protocols. Utilizing our UAV-swarm-aided DRP implementation, we conduct experiments with real-world UAS traces, encompassing diverse mobility models, geospatial link information, and video analytics measurements. This research not only advances the understanding of UAV simulation capabilities but also emphasizes a user-centric approach through the incorporation of HCI concepts in DRP design.

The remainder of the paper is organized as follows: In Sect. 2, we describe the UAV-swarm-aided disaster response platform design in detail. Section 2.2 details our model strategies related to video transmission and coordination to achieve air-to-ground communication. Section 4 describes the platform experiment results and salient findings. Lastly, Sect. 5 concludes the paper.

2 UAV-Swarm-Aided DRP Design and Modeling

In this section, we initiate our exploration by offering a comprehensive overview of our platform, meticulously outlining each component intricately linked to its functionality. To ensure a thorough analysis and to present a holistic view of our output processes, we delve into the specifics of air-to-ground video transmission and coordination modeling, providing detailed descriptions of the design and operational intricacies.

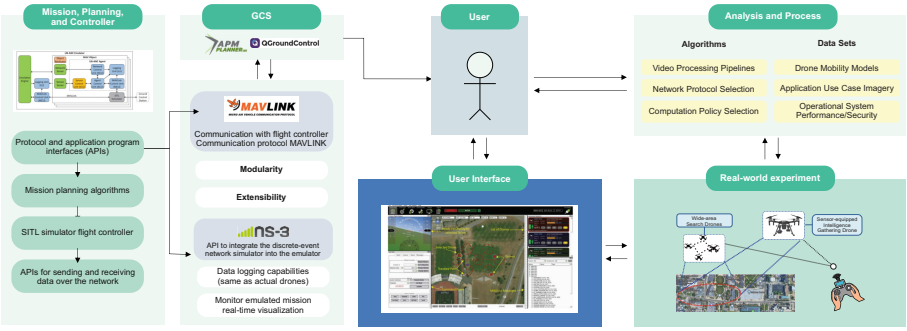


Fig. 2. User has interaction with the components of the Air-to-Ground Coordinator. Analysis and processes are done based on the outputs from the video transmission. The emulator is integrated with a real-world experiment component, and its functionality should include the emulation of any MAVLink-compatible vehicle.

2.1 Design Overview

Figure 2 provides an overview of our proposed solution approach, specifically focusing on an Air-to-ground Coordination scenario within Disaster Response Processes (DRP). Our approach incorporates *system* awareness (e.g., energy levels), *mobility* awareness (e.g., ground sensor location, UAV trajectory), and *environmental* awareness (e.g., physical obstacles, wind intensity) modeling. This comprehensive awareness modeling guides learning-based video transmission and offloading strategies intelligently and jointly.

In our hierarchical UAV system connected to the edge, we deploy both low-capability search UAVs, covering extensive geographic areas, and high-capability intelligence gathering UAVs for fine-grained surveillance in targeted regions. These UAVs can be deployed both in the field and within a network-embedded UAV simulator. Human operators effectively control UAV swarms through a single orchestration end-point, enabling interactive and coordinated operations among the UAV swarms for streamlined management.

Moreover, the synergistic operation of UAVs and network emulators offers benefits to human operators, including the ability to (a) test various system configurations on the networking side and (b) implement different flight algorithms before deploying them in disaster-stricken areas. This approach enhances the efficiency and adaptability of disaster response operations through the integration of intelligent UAV swarms and network emulation capabilities.

In conjunction with the human-centered interaction graphical user interface (GUI) design, our approach incorporates open-source API design to establish connectivity with the NS-3 network simulation platform, as documented in [11]. The integration of Mission, Planning, and Controller components facilitates the dissemination of detailed flight information to GCS operators, enabling them to make informed pre-flight decisions. Throughout the take-off phase, operators can leverage our proposed user interface system, outlined in Fig. 2, to perform

additional operations. Notably, the user interface supports the concurrent operation of multiple drones within a single page, with each interaction manifesting in real-world drone flights. Furthermore, the system captures and logs datasets, storing them in files for subsequent analyses and operations.

2.2 Air-to-Ground Video Transmission and Coordination Modeling

In this section, we detailed our video transmission approach between air-to-ground, considering various video transmission protocols and network setups. After that, we discuss the coordination of drones and GCS to provide a better transmission environment by briefly introducing the orchestration between multi-drones and GCS in the disaster response management scenario.

Video Transmission Modeling: In response to user specifications, we will organize diverse solutions into categorized lists, addressing various aspects. The categories encompass a range of protocols, video codecs, and video resolutions. Each category comprises distinct options, including (i) Network protocols, offering choices among HTTP, TCP, UDP, RTP, and QUIC; (ii) Video Properties, with a selection between H.265 HEVC (High-Efficiency Video Coding) and H.264 AVC (Advanced Video Coding) for video codec; and allowing choices between 720p, 1080p, and 2K for video resolution.

To enhance predictive modeling, we correlate Application-layer and Transport-layer Protocols with video transmission properties, encompassing network protocol, video codec, and video resolution. In practical scenarios, the unpredictable wireless environment poses challenges in precisely anticipating network conditions during drone flights. Furthermore, rapid changes in network protocols aligned with environmental situational awareness requirements may occur along the flight paths. Consequently, selecting network protocols and video properties at the commencement of a drone flight assumes paramount importance. Given the absence of detailed information on network and video codec selection in the comprehensive dataset of detailed video transmissions captured by drones, we conducted additional video transmission tasks within our proposed user interface platform. Specifically, for each trace obtained from the dataset, varied combinations of network and video properties were employed to generate outputs, such as 1080p video encoded with H.265 under HTTP/TCP communication, and subsequent observations of the results were made. The flowchart depicted in Fig. 3 delineates the decision-making process employed in the selection of network protocols and video properties. The evaluation of the modeling incorporates both Peak Signal-to-Noise Ratio (PSNR) as an objective criterion and Mean Opinion Scores (MOS) as a subjective criterion. Comprehensive details of the evaluation results are available in Sect. 4.3.

Air-to-Ground Coordination Modeling: To streamline these processes, we bifurcate collaboration and orchestration into two distinct phases: the pre-application analysis phase and the real-time in-application prediction phase. The initial step involves gathering traces from diverse scenarios under both Disaster response (DR) setups. For each DR scene, we execute benchmark video streaming

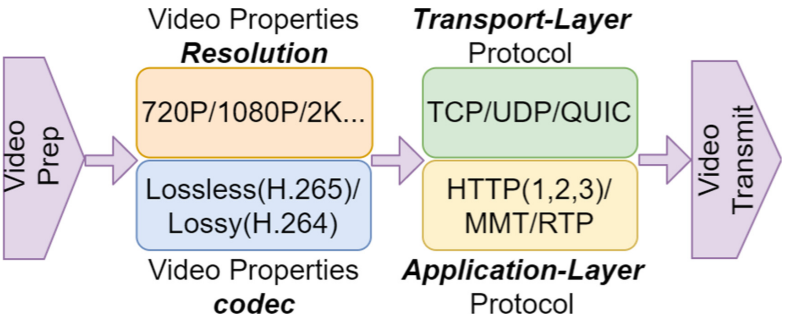


Fig. 3. Flowchart detailed the process of selecting network protocols and video properties using our GUI platform. The GUI provides detailed output on the video component, supporting multiple UAV video transmissions with the same configurations.

analytics to simulate performance across three execution categories: local execution (on a single UAV), edge server execution (complete resource offloading onto the edge server), and UAS-swarm-Server collaboration execution (leveraging joint learning-based computation offloading approach [3] for task distribution). The benchmarking encompasses network protocols and video properties as detailed in the previous section. Thresholds, derived from DR scene analysis results, factor in crucial metrics like recognition accuracy, map-generating rate, and DR scene response time. The outcomes of policy-based decisions are then applied to each UAV for subsequent deployment.

While the utilization of benchmark analysis results optimizes performance and facilitates policy-based decisions in the UAV-swarm-aided with Edge server, i.e., air-to-ground collaboration, there arises a necessity to dynamically orchestrate the operational processes by predicting traces and making advanced decisions during operations. In response, we propose a Deep Q-Network (DQN)-based trajectory prediction algorithm, generating predictive traces for each moving UAV. This DQN trajectory prediction algorithm is applied to each UAV involved in task sharing across various mobility models [5], such as random waypoints, Gauss-Markov, and mission-plan-based. The precise trajectory predictions are distributed for each DR scene, and decisions are guided by policy-based collaboration strategies. To implement this, each *wide-area search UAV* and *multi-sensor intelligent gathering UAV* embed (as described in the right-hand side of the Fig. 2) reference weights from the learning output of the DQN algorithm. To enhance the prediction procedure, we compare several DQN methods, including DQN, Double-DQN, and Duelling-DQN. The evaluation output (see Sect. 4.3), encompassing network performance and overall gains, is discernible across the various DQN methods listed.

3 UAV-Swarm-Aided Disaster Response Platform User Interface Design

In this section, we describe the user interface design of our proposed UAV-swarm-aided DRP. In detail, we delve into the detailed description of the open-sourced user interface, highlighting components related to human-centered interaction strategies. Finally, we introduce the software components and dependencies that have been employed in designing the platform.

3.1 User Interface Design and HCI Components

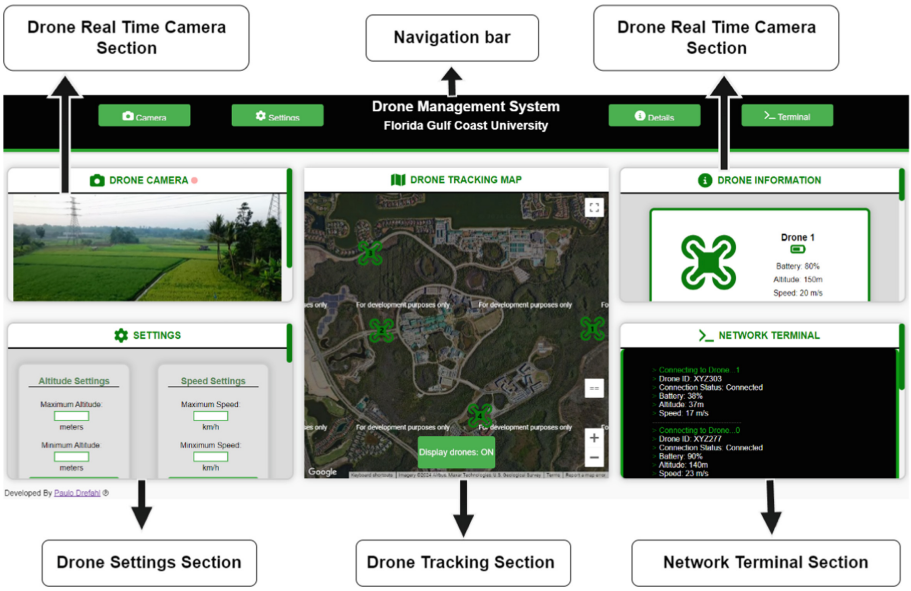


Fig. 4. Snapshot of the initial dashboard in the DRP Graphical User Interface (GUI), Highlighting Key Elements and Functionalities. Exploration of all the techniques and SPA design patterns, including (from left to right) i) drone cameras, ii) navigation bar, iii) drone information details, iv) general settings, v) tracking map, and vi) network terminal.

In the realm of UAV management, the integration of HCI principles and advanced GUI design is paramount for elevating user experience and ensuring system usability across diverse devices. Detailed analysis of Fig. 4 spotlighting the distinctive features of the software UI and its design patterns, with a particular emphasis on responsiveness, clarity, and adaptability. These qualities collectively contribute to an efficient and user-friendly interface tailored for controlling and monitoring drones.

A significant challenge associated with software of this nature, dealing with substantial amounts of UAV-related data and variables, lies in presenting this information in an accessible and user-friendly manner. Constantly exploring information visualization concepts [15]. The data must be readily available for monitoring by the GCS Agent or individuals using the software, ensuring UAVs are performing optimally and extracting valuable insights from the received information. The first design pattern that emerges is the *Single Page Application (SPA)* [14], coupled with a *modal* or *"lightbox"* technique.

SPA consolidates all pertinent information onto a single webpage, eliminating the need for users to navigate through multiple pages. This approach provides a comprehensive view of i) drone cameras, ii) navigation bar, iii) drone information details, iv) general settings, v) tracking map, and vi) network terminal. By optimizing resource loading and dynamically updating content, SPAs significantly reduce loading times, ensuring quick and efficient access to critical information [14]. In conjunction with SPA, the incorporation of Modal Design Patterns further enhances the user interface. When users interact with specific elements, such as a drone camera or network terminal, a modal overlay extends the relevant section, offering detailed information, charts, or controls related to that drone. This design pattern enables users to delve into specific details without navigating away, fostering a more immersive and efficient interaction.

These interactions are made possible by various UI elements seamlessly integrating HCI principles to ensure optimal user experience and system operability across diverse devices. At the top of Fig. 4, the Navigation Bar presents clear icons and labels, facilitating intuitive interaction. Recognizable buttons enhance user navigation, focusing on specific sections when pressed. This design, responsive across platforms, allows for effortless navigation through various functionalities, aligning with the SPA pattern.

The primary drone information is housed in the Drone Information Box at the top right corner of Fig. 4. It serves as a user-friendly display, emphasizing easy comprehension of drone details such as battery status, drone ID, altitude, coordinates, etc. Hover-over tooltips and touch interactions on mobile devices enable quick exploration. Responsive design ensures a seamless experience, expanding upon interaction to provide users with a comprehensive view of all connected devices. This aligns with the SPA pattern, consolidating information for a unified and accessible display.

Crafted for diverse devices, the Settings Interface within the bottom right corner of Fig. 4 allows users to customize drone parameters effortlessly. Whether adjusting sliders on a computer or using touch controls on mobile devices, the design ensures optimal interaction and adaptability, reflecting the SPA pattern's responsiveness. The expandable settings box accommodates varying input requirements, enhancing the precision of user inputs and aligning with the Modal Design Pattern.

To track drones, a Drone Tracking Map was implemented, prominently displayed at the center of the application (Fig. 4). It facilitates interactive HCI and integrates drone locations using an API from Google Maps and a real-time

server that retrieves coordinates from the drones. The ability to manipulate the map through a mouse or touch gestures on mobile devices ensures user-friendly navigation. Responsive design empowers users to actively control drone icons, fostering situational awareness. Features such as map navigation, linking drones to the information box, and toggling drone locations align with the SPA and Modal Design Patterns. Functioning as a system log display, the Network Terminal Box ensures transparent HCI. Located at the bottom right of the page (Fig. 4), responsive design and clarity in presenting information make it suitable for effective monitoring and troubleshooting. Users can scroll through logs on computers, and touch interactions facilitate navigation on iPad, allowing the terminal to expand and scroll through even on mobile devices. The expandable terminal box provides a comprehensive view of real-time updates from the drones, demonstrating a seamless integration of HCI principles and UI design elements. This UAV-swarm-aided DRP is open-source and can be downloaded and inspected at the GitHub repository [4].

3.2 Software Components and Dependencies

In terms of the software components and dependencies used to develop this DRP, each component has undergone a rigorous requirements-gathering process and multiple design iterations to facilitate the Human-Computer Interaction (HCI) aspects, providing operators with an intuitive platform for efficient drone oversight and control. The front-end utilizes a combination of HTML, CSS, JavaScript, and complementary frameworks such as Bootstrap. This amalgamation forms the backbone of the software's user interface, enabling seamless access and navigation through various sections, as highlighted in the Navigation Bar (see Table 1). Bootstrap enhances the responsiveness of the navigation bar, aligning with the software's commitment to optimal user experience.

The Drone Camera Box, a focal point of the user interface, leverages several technologies to deliver a rich and responsive experience. Real-time video communication is achieved through WebRTC [7], utilizing a back-end API in JavaScript to handle protocols for receiving live video feeds from drones. The HTML5 video element and JavaScript animations contribute to toggling functionalities and displaying a recording indicator. This integration aligns with the software's emphasis on clear communication of the camera's status.

The Drone Tracking Map Box, crucial for real-time drone location monitoring, harnesses the power of the Google Maps API along with JavaScript, HTML, and CSS. This combination allows the presentation of real-time drone positions on an interactive map. Animations enhance user engagement, facilitating intuitive navigation through the map and dynamic control of drone display. This technological synergy, outlined in Table 1, underscores the software's scalability and its ability to enable real-time tracking possibilities in the future.

To monitor real-time updates about individual drones, a Drone Information Box has been developed, facilitated by HTML, CSS, and JavaScript. The seamless integration of these technologies ensures a user-friendly display of critical information, promoting easy comprehension across devices. Information requests

Table 1. Overview of System Components with Descriptions and Main Functionalities. All are displayed in SPA pattern with boxes that are dynamically extended based on Ground Control Operators (users) interaction.

Component	Description
Frontend Application	User interface for interacting with the drone management system.
Navigation Bar	Navigation bar for easy access to different sections of the software.
Drone Camera Box	Displays live video feed from the drone camera, toggle functionality, and recording indicator.
Drone Tracking Map Box	Shows a map with real-time drone positions, toggle display button.
Drone Information Box	Displays real-time information about individual drones.
Settings Box	Allows users to configure various drone parameters such as altitude, speed, rotation, and camera settings.
Network Terminal Box	A terminal-like box for displaying system logs and messages.
Animation Script	Responsible for animating drones on the map and providing dynamic visual feedback.
Backend node.js	Manages data, handles user requests, and communicates with drones and external systems.
Stylesheets	CSS files for styling the user interface.
Drone Communication	Integration with drones for real-time data exchange

ping through an API in the Node.js backend at specific intervals to ensure the accuracy of the displayed data. Hover-over tooltips and touch interactions enhance the responsiveness of this component, contributing to an efficient and transparent operational environment.

In software development, the ability to interact with settings and platform specificity is crucial. Hence, a Settings Box has been added, designed for configuring various drone parameters, utilizing HTML, CSS, and JavaScript. This technology functions in synchrony with the Network Terminal Box. Together, they allow users to input and handle configurations with ease while observing real-time responses and updates from the drones.

In summary, the technological components detailed in this section form a robust foundation for the DRP user interface. Through meticulous requirements gathering and iterative design, the front end employs a blend of HTML, CSS, JavaScript, and supportive frameworks like Bootstrap. This amalgamation serves as the backbone, ensuring an intuitive platform for Ground Control Operators to oversee and control drones efficiently.

4 Platform Experiment Results and Findings

In this section, we first provide the experiment setups with both simulation and real-world setups. To evaluate our performance in the DRP, we provide several experiments under both simulator and real-world setups considering various metrics in terms of time efficiency and energy-saving. Objective and subjective results related to video transmission are also provided to evaluate the human-center interaction principles.

4.1 Experiment Setups

In this experiment setups, we consider both simulator and real-world experiments. To emulate the streaming process among UAVs and the Edge server, we utilized FFmpeg to systematically configure raw data into videos, employing different codecs (h.264, h.265) and setting various video resolutions (720×480, 1280×720, 1920×1080), all at a constant 25 FPS. This effort resulted in the generation of 100 distinct videos. Subsequently, these videos were executed on a Nvidia Jetson Xavier NX using the YoloV8 [17] model to perform real-time object detection. The output included key confidence data and processing speed. In the context of our network performance evaluation, we implemented a network offloading strategy for drone swarms utilizing TCP video streaming. The streams were limited by different bit rates (500kbs, 1mbs), resulting in ≈ 500 unique streaming situations.

In terms of the real-world experiments, we conducted data collection from a multi-hierarchical drone configuration using centralized-control communication networks, considering various mobility models and varying fleet sizes. Each drone is operated by one Ground Control Station (GCS), and all traces and computation processes are centrally controlled by a single edge server. To streamline our experiments, we focused on two aspects of data: drone video capture data and drone trace data.

For drone video capture data in the real-world experiment category, we gathered 30 video clips from three types of drones (two Skydio 2+ drones, two Parrot Anifa AI drones, and one Parrot Anifa USA drone). Each drone had three different mobility settings: mission-based plan model, Gaussian-Markov model, and random model. Drone video clips were transmitted to GCS via a simple, reliable, and widely available data link, such as IEEE 802.11. Metrics were calculated based on both video codec type and video resolution.

In terms of drone flight trace gathering, we utilized real-world experiments to generate fleets of 2 and 4 drones. Traces were formatted as JSON files and stored in a NoSQL database. Metrics were calculated on (i) the goodput through a network monitor at a GCS and (ii) detailed flight traces information, including longitude, height, speed (mph), and ascent (feet). For every millisecond, one throughput data point was sent from the drones to the GCS for recording and archival purposes.

4.2 Evaluation Metrics Discussion

In evaluating video transmission results, a comprehensive assessment involves both subjective and objective metrics. Subjective evaluation, conducted through a Mean Opinion Score (MOS) questionnaire, captures users' perceptions and experiences across key categories such as visual presentation, sound quality, smooth playback, absence of freezes, and overall satisfaction. MOS provides a qualitative measure, allowing users to express their opinions and preferences on the multimedia content. This human-centric approach is crucial for understanding the real-world impact of video transmission on user satisfaction. Detailed topics, example questions, and the number of questions list can be found in Table 2.

On the other hand, incorporating objective metrics is essential for a more technical and quantitative assessment. Peak Signal-to-Noise Ratio (PSNR) serves as an objective metric, measuring the quality of the transmitted video signal by evaluating the difference between the original and received signals. PSNR offers a numerical representation of video fidelity, providing insights into the technical performance of the transmission.

The synergy between subjective MOS evaluations and objective PSNR metrics is beneficial for a holistic understanding of video transmission quality. While MOS captures the user's perceptual experience, PSNR quantifies the technical aspects, ensuring a well-rounded evaluation. Combining both subjective and objective assessments enhances the robustness of the evaluation process, facilitating a comprehensive analysis that considers both user satisfaction and technical quality. This dual approach is pivotal for addressing diverse aspects of video transmission, ultimately leading to more informed decisions and improvements in multimedia delivery systems.

Table 2. Questionnaire detailed topics, example questions, and the number of questions list for video transmission results.

Catgory	Topic	Question Example - No. of Questions
1	Visual Presentation	How satisfied with video clarity? (1-5) - 5
2	Sound Quality	How clear is the audio? (1-5) - 3
3	Smooth Playback	Rate video smoothness (1-5) - 2
4	Absence of Freezes	Any interruptions or freezes? (1-4) -3
5	Effectiveness	Satisfied with absence of disruptive elements? (1-5) - 2
6	Other Issues	Notice any concerns? Specify. −1
7	Overall Satisfaction	Rate overall satisfaction (1–5). - 1

The evaluation metrics for our HCI user interface and Air-to-Ground Coordination system again encompass both subjective and objective measures to ensure a comprehensive understanding of performance. To evaluate the user interface experience, we rely on feedback from GCS agents, treating their opinions as subjective metrics. Our open-source UI, freely accessible on community discussion

boards and regularly updated logs on GitHub, encourages transparency and collaboration, fostering a participatory environment. We have specifically engaged Missouri Soybean field farmers and Florida citrus farmers, providing them with open access to the UI and collecting their opinions to enhance usability.

In terms of objective metrics, we focus on network performance and energy efficiency. Network performance is evaluated through latency and end-to-end delay, measuring the operational efficiency of communication between the GCS agent and the drone. This ensures that real-time data transmission is efficient and responsive. For air-to-ground coordination, we consider optimal scheduling time and energy consumption rate. The former ensures effective coordination timing, optimizing the communication between the air and ground components. Meanwhile, the latter quantifies energy usage for coordination relative to energy usage for flight, providing insights into the system's overall energy efficiency.

The ongoing collection of opinions from farmers reflects our commitment to iterative improvement based on user feedback. The combination of subjective and objective metrics in our evaluation approach ensures a well-rounded understanding of the HCI and Air-to-Ground Coordination system, paving the way for continual enhancements to meet user needs and optimize system performance.

4.3 Evaluation Experiments and Findings

Table 3. Various experiment comparison results (subjective: MOS and objective: PSNR) in terms of the video transmission. Comparison methods include real-world experiments (using built-in video transmission), SITL experiments [10] (based on simulator and traces), DroneNet-Sim [11], our proposed platform, and oracle (original video source).

Experiments	MOS							PSNR
Real-World	3.5 ± 0.75	2.5 ± 0.5	1.9 ± 0.25	1.8 ± 0.5	1.6 ± 1.5	1.8 ± 0.6	2.1 ± 0.8	28 ± 5
SITL	2.3 ± 0.75	3.2 ± 0.5	3.0 ± 0.2	3.2 ± 0.5	3 ± 0.27	3.5 ± 0.7	3.8 ± 0.3	26 ± 10
DroneNet-Sim	3.5 ± 0.6	4.2 ± 0.2	4.0 ± 0.15	4.1 ± 0.1	4.2 ± 0.5	**4.5 ± 0.2**	4.3 ± 0.3	30 ± 5
Ours	**4 ± 0.5**	**4.8 ± 0.2**	**4.5 ± 0.3**	**4.8 ± 0.2**	**4.6 ± 0.2**	4.3 ± 0.1	**4.5 ± 0.5**	**40 ± 10**
Oracle	4.5 ± 0.5	4.7 ± 0.25	4.8 ± 0.12	4.7 ± 0.15	4.4 ± 0.6	4.8 ± 0.1	4.6 ± 0.35	N/A

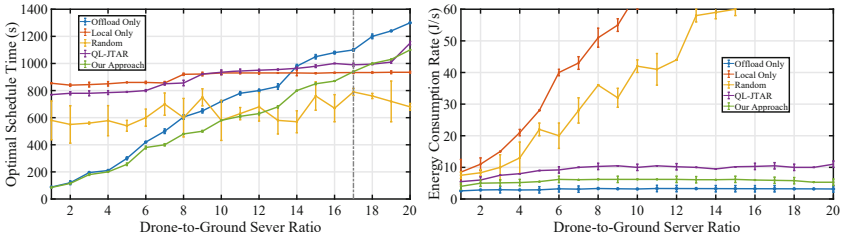
Table 3 presents a comparative analysis of various image processing methods, including our own, evaluated through Mean Opinion Score (MOS) and Peak Signal-to-Noise Ratio (PSNR) metrics across diverse disaster response scenarios. As we can observe from the results, our method consistently outperforms Real-World, SITL, and DroneNet-Sim counterparts considering both MOS and PSNR values. This superiority is evident not only in terms of higher subjective and objective image quality but also in showcasing lower variability. Across diverse scenarios, our method demonstrates an overall MOS measurement of 4.5 ± 0.5,

surpassing Real-world (2.1 ± 0.8), SITL (3.8 ± 0.3), and DroneNet-Sim (4.3 ± 0.3). In terms of the objective metrics PSNR, our video transmission output again outperformance other comparison methods with an overall of 35–45 PSNR that highlights the robust and consistent empirical performance of our method. Notably, our approach closely approaches the Oracle with minimal differences, indicating image quality on par with the theoretical best in the evaluated context. In conclusion, the proposed method excels in both perceived and measured image quality, surpassing counterparts and approaching the optimal quality represented by the Oracle.

The evaluation results presented in Fig. 5 illustrate the performance of our air-to-ground coordination solution implemented in the Air-to-Ground Coordinator User interface, as detailed in the Method section. Based on the collected data, we draw the following conclusions:

Observation 1: Our coordination scheme, leveraging air-to-ground coordination, demonstrates superior performance compared to local execution and the Q-learning-based QL-JTAR [3] solution, particularly in scenarios with low Air-to-Ground ratios. Figure 5a highlights that our approach aligns with Offload Only, where human operators fully coordinate, resulting in a noteworthy improvement of up to 15% in scheduling make-span compared to offload-only schemes (and achieving ≈ 85% accuracy with dynamic decision-making). Random offloading exhibits erratic behavior with significant variance, while the scheduling time remains consistently optimal for exclusive local UAV processing, preventing ground queue buildup. In contrast, QL-JTAR shows inefficiency, requiring approximately 800 s in all experiments, particularly proving ineffective for low ratios. Our experimental focus on low Air-to-Ground ratios underscores the excellence of our approach in such scenarios.

Observation 2: Our air-to-ground coordination approach surpasses Random offloading and Local Only execution in terms of energy consumption, providing considerable energy savings (approximately 18% gain) compared to onboarding only, in addition to offering reasonable energy savings relative to the QL-JTAR solution. As demonstrated in Fig. 5b, the energy consumption rate sharply increases with rising Air-to-Ground ratios for Local Only and Random offloading strategies. However, our approach consistently maintains a lower energy consumption rate than these methods, albeit slightly higher than the Offload Only scheme. This discrepancy is justifiable as the Offload Only scheme only considers transmission energy, excluding video processing energy. Nevertheless, the energy difference between our approach and the Offload Only scheme remains relatively small. Considering our approach's up to 40% improvement in scheduling time over the Offload Only scheme, it becomes evident that our method excels when both time and energy considerations are taken into account.



(a) (b)

Fig. 5. Various comparisons of air-to-ground coordination strategies have been conducted, encompassing Offload Only (fully coordinated by human operators), exclusive local execution (UAV only), a random approach, the QL-JTAR learning-based approach, and our air-to-ground coordination approach. The results are presented in terms of (a) the correlation between increasing Air-to-Ground ratio (UAV swarm size: human operator) and optimal schedule time (in seconds) and (b) the relationship between Air-to-Ground ratio and the average energy consumption per UAV (in Joules).

5 Conclusion

In this paper, we discuss the role of UAV-swarm-aided in diverse applications and their impact on environmental situational awareness, the complexities associated with constructing UAV testbeds, and highlights the difficulties in achieving repeatability and reproducibility. To address these challenges, we investigate state-of-the-art approaches in advanced UAV simulators, proposing a trace-based simulation design utilizing NS-3. The hierarchical UAV-swarm-aided platform, involving collaboration between low-cost and high-cost drones, demonstrates the synergistic benefits of integrating network and drone simulators. The implemented simulator, designed with Human-Computer Interaction (HCI) principles, advances the understanding of UAV simulation capabilities and emphasizes a user-centric approach. Based on the experiments with real-world UAV traces, encompassing diverse mobility models and video analytics measurements, our platform not only contributes to advancing UAS technology but also prioritizes user experience and system usability in the design of simulators. Futures works can be

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